Hartree-Fock Method Summary

Irma Syafitri
Personal Summary
Applied Physics, Osaka University

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1 Introduction

Here, I'd like to share my study notes on the **Hartree-Fock method** (under the Born-Oppenheimer approximation). Why do we even need this approach in the first place? Well, if we have two or more electrons and want to calculate their total energy, solving the Schrödinger equation directly gets ridiculously complicated. That's why we need a special approximation method, and in these notes, we're going to use Hartree-Fock.

Naturally, we'll be using calculus here, since calculus is basically the language of physics. Before we dive into the physical phenomena, let's jump right in!

$$E_{\text{HF}} = T_e + V_{ne} + E_{\text{Coulomb}} + E_{\text{exchange}}$$

where T_e is the kinetic energy of the electrons, V_{ne} is the nucleus-electron attraction, E_{Coulomb} is the classical electron-electron repulsion, and E_{exchange} is the quantum exchange interaction.

2 Hamiltonian operator

Before we can break down the total energy, we need to look at the main operator: the **Hamiltonian operator ($\hat{H}$)**.

For a system with N electrons and a single nucleus (under the Born-Oppenheimer approximation, where the nucleus stays put):

$$\hat{H} = \sum_{i=1}^{N} \left[-\frac{\hbar^2}{2m} \nabla_i^2 - \frac{Ze^2}{4\pi\epsilon_0 r_i} \right] + \sum_{i<j}^{N} \frac{e^2}{4\pi\epsilon_0 r_{ij}} \tag{1}$$

Briefly breaking down the terms:

- $\sum_{i=1}^{N} \left(-\frac{\hbar^2}{2m} \nabla_i^2 \right)$: The kinetic energy ($T_e$) of all N electrons.
- $\sum_{i=1}^{N} \left(-\frac{Ze^2}{4\pi\epsilon_0 r_i} \right)$: The attractive potential between the nucleus and the electrons (V_{ne}).

- $\sum_{i<j}^{N} \frac{e^2}{4\pi\epsilon_0 r_{ij}}$: The classical repulsion between electron pairs (E_{Coulomb}), this is the exact reason why solving the exact Schrödinger equation is a nightmare!

2.1 The Electron-Repulsion Breakdown: J_{ij} and K_{ij}

Now, if we look closer at that tricky electron-electron repulsion term, this is where things get really interesting. In the Hartree-Fock method, we break down this interaction into two main parts: J_{ij} **(the Coulomb term)** and K_{ij} **(the Exchange term)**.

- J_{ij} **(Coulomb integral):** This represents the classical electrostatic repulsion between two electron density clouds (averaging out where the electrons are).
- K_{ij} **(Exchange integral):** This is a purely quantum mechanical effect with no classical equivalent. It arises entirely due to the Pauli exclusion principle for electrons with parallel spins, keeping them safely apart!

3 The Coulomb Integral (J_{12})

Let's start by looking closely at the Coulomb term for a two-electron case:

$$J_{12} = \iint \chi_1^*(1)\chi_2^*(2)\frac{1}{r_{12}}\chi_1(1)\chi_2(2)\, d1\, d2$$

Remember that each spin-orbital $\chi_i(x)$ consists of a spatial orbital $\phi_i(\mathbf{r})$ and a spin function $\omega(\sigma)$:

$$\chi_i(x) = \phi_i(\mathbf{r})\omega(\sigma)$$

Substituting this into our integral and grouping the spatial and spin coordinates (knowing that total volume element $dx = d\mathbf{r}d\sigma$):

$$J_{12} = \left[\int \omega_1^*(\sigma_1)\omega_1(\sigma_1)\, d\sigma_1\right]\left[\int \omega_2^*(\sigma_2)\omega_2(\sigma_2)\, d\sigma_2\right] \times \iint \frac{|\phi_1(\mathbf{r}_1)|^2|\phi_2(\mathbf{r}_2)|^2}{r_{12}}\, d\mathbf{r}_1 d\mathbf{r}_2$$

Since the spin functions are normalized, both spin integrals equal 1:

$$\int \omega_1^*\omega_1\, d\sigma_1 = 1 \quad \text{and} \quad \int \omega_2^*\omega_2\, d\sigma_2 = 1$$

Therefore, we arrive at the spatial Coulomb integral:

$$\boxed{J_{12} = \iint \frac{|\phi_1(\mathbf{r}_1)|^2|\phi_2(\mathbf{r}_2)|^2}{r_{12}}\, d\mathbf{r}_1 d\mathbf{r}_2}$$

If we define $\rho_1(\mathbf{r}_1) = |\phi_1(\mathbf{r}_1)|^2$ as the probability density of electron 1 and $\rho_2(\mathbf{r}_2) = |\phi_2(\mathbf{r}_2)|^2$ as that of electron 2, we can rewrite it as:

$$\boxed{J_{12} = \iint \frac{\rho_1(\mathbf{r}_1)\rho_2(\mathbf{r}_2)}{r_{12}}\, d\mathbf{r}_1 d\mathbf{r}_2}$$

Simply put, J_{12} is physically identical to the "classical electrostatic repulsion between two charge distributions"!

3.1 The Exchange Integral (K_{12})

Now, let's look at the more intriguing part: the "exchange term (K_{12})". Unlike the Coulomb term, the electron coordinates on the right side are swapped:

$$K_{12} = \iint \chi_1^*(1)\chi_2^*(2)\frac{1}{r_{12}}\chi_2(1)\chi_1(2)\,d1\,d2$$

Using the substitution $\chi_i = \phi_i\omega_i$ and grouping the spatial and spin coordinates, we get:

$$K_{12} = \left[\int \omega_1^*(\sigma_1)\omega_2(\sigma_1)\,d\sigma_1\right]\left[\int \omega_2^*(\sigma_2)\omega_1(\sigma_2)\,d\sigma_2\right] \times \iint \frac{\phi_1^*(\mathbf{r}_1)\phi_2(\mathbf{r}_1)\phi_2^*(\mathbf{r}_2)\phi_1(\mathbf{r}_2)}{r_{12}}\,d\mathbf{r}_1 d\mathbf{r}_2$$

Here is where spin determines whether the exchange term survives or vanishes:

- If the spin functions are different ($\omega_1 \neq \omega_2$), due to spin orthogonality, the spin integrals equal 0, which means:

$$K_{12} = 0 \quad \text{(for opposite/different spins)}$$

- If the spin functions are the same ($\omega_1 = \omega_2$, e.g., both α or both β spins), the spin integrals equal 1, leaving us with the spatial exchange integral:

$$K_{12} = \iint \frac{\phi_1^*(\mathbf{r}_1)\phi_2(\mathbf{r}_1)\phi_2^*(\mathbf{r}_2)\phi_1(\mathbf{r}_2)}{r_{12}}\,d\mathbf{r}_1 d\mathbf{r}_2 > 0 \quad \text{(for parallel spins)}$$

This proves that the exchange energy only acts between electrons with parallel spins, keeping them physically separated as dictated by the Pauli exclusion principle!

4 Case Study: Boron and Carbon Atoms

To make things more concrete, let's look at specific atoms using our notation. From Equation (1), we can define the "one-electron operator $h(i)$" as:

$$h(i) = -\frac{\hbar^2}{2m}\nabla_i^2 - \frac{Ze^2}{4\pi\epsilon_0 r_i}$$

which groups the electron's kinetic energy and its attraction to the nucleus. Meanwhile, the two-electron part yields our Coulomb (J) and Exchange (K) terms, where the minus sign in $J - K$ fundamentally originates from the antisodicity of the Slater determinant (Pauli exclusion principle).

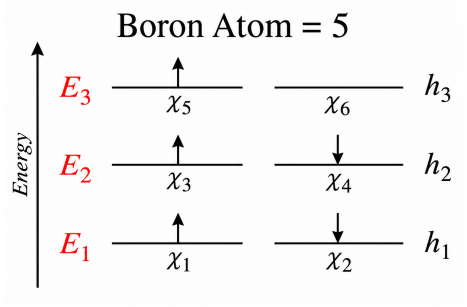


Figure 1: Energy-level diagram for a Boron atom ($Z = 5$).

4.1 Energy Calculation for Boron Atom ($Z = 5$)

For a Boron atom with 5 electrons, the total Hartree-Fock energy accounts for the core and valence orbitals, summing up the one-electron core. Hamiltonians and the pairwise $(2J - K)$ electron-electron interactions. The Restricted Hartree-Fock total energy (E_{RHF}) can be expressed as (Figure 1):

$$\begin{aligned} E_{\text{RHF}} = {} & 2h_1 + 2h_2 + h_3 \\ & + J_{11} + 4J_{12} - 2K_{12} + 2J_{13} - K_{13} \\ & + J_{22} + 2J_{23} - K_{23} \end{aligned} \tag{2}$$

4.2 Energy Calculation for Carbon Atom ($Z = 6$)

Moving to Carbon with 6 electrons, we add another valence electron into the $2p$ subshell, introducing additional intra-shell Coulomb and exchange interactions among the $2p$ electrons.

Based on the energy-level diagram where the 6 electrons occupy orbitals χ_1 through χ_6 (from Figure 2 grouped into spatial orbitals h_1, h_2, h_3, h_4), the total Restricted Hartree-Fock energy (E_{RHF}) is expressed as:

$$\begin{aligned} E_{\text{RHF}} = {} & 2h_1 + 2h_2 + h_3 + h_4 \\ & + J_{11} + 4J_{12} - 2K_{12} + 2J_{13} - K_{13} + 2J_{14} - K_{14} \\ & + J_{22} + 2J_{23} - K_{23} + 2J_{24} - K_{24} \\ & + J_{34} - K_{34} \end{aligned} \tag{3}$$

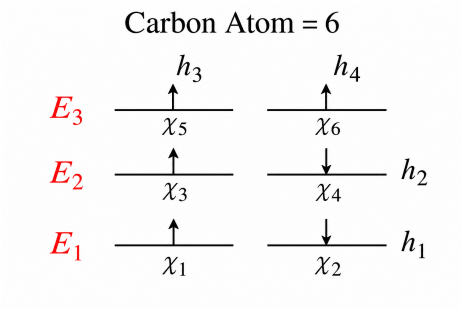


Figure 2: Energy-level diagram for a Carbon atom ($Z = 6$).

5 Results and Discussion: Effect of Atomic Number on Total Energy

From the Quantum ESPRESSO calculations, we obtain the total energies for Boron ($Z = 5$) and Carbon ($Z = 6$) as summarized in Table 1.

Table 1: Total energy comparison between Boron and Carbon atoms.

Atom	**Atomic number (Z)**	**Total energy (Ry)**
Boron (B)	5	-5.42412204^*
Carbon (C)	6	-10.77572728

Observed trend: The energy of carbon is more negative than the energy of boron.
Difference in total energy:

$$\Delta E = E_{\mathrm{C}} - E_{\mathrm{B}} = -5.35160524 \text{ Ry} \tag{4}$$

$$E_{\mathrm{C}} < E_{\mathrm{B}} \tag{5}$$

The calculated total energy becomes more negative as the atomic number increases from boron ($Z = 5$) to carbon ($Z = 6$). Increasing the atomic number increases the nuclear charge, which strengthens the attractive interaction between the nucleus and electrons.

However, adding an electron also changes electron–electron repulsion, kinetic energy, and the electronic distribution. Therefore, the total energy is determined by the balance of these contributions.

For the calculated neutral B and C atoms, the results show $E_{\mathrm{C}} < E_{\mathrm{B}}$, indicating that the total energy of carbon is lower than that of boron under the computational conditions used.